

Section: 03	Group: 03	Status:
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Table 1.1: Component Values

	$R_1(\Omega)$	$R_2(\Omega)$	$C_1(F)$	$L_1(H)$
Nominal	100 Ω	200 Ω	0.1 μF	330 μH
Measured	100.4 Ω	200.1 Ω	0.11 μF	317.3 μH

Table 1.2: The magnitudes of V_R at different frequencies for a high Q circuit

Frequency (f) kHz	Peak Voltage, $V_R(V)$	$I_{peak}(mA)$	
1	240 mV	2.39 mA	
10	1.76 V	17.5 mA	
20	2.76 V	27.5 mA	
25	2.88 V	28.69 mA	
27.7	2.92 V	29.08 mA	
30	2.92 V	29.08 mA	
40	2.80 V	27.89 mA	
50	2.60 V	25.9 mA	
70	2.12 V	21.12 mA	
100	1.60 V	15.94 mA	
120	1.36 V	13.55 mA	

Table 1.3: Resonant frequency, Quality Factor and Bandwidth for a high Q circuit

	Theoretical, (a)	Experimental, (b)	Deviation (%), $\frac{a - a}{b} \times 100$
f₀ (KHz)	27.7 kHz	27.7 kHz	0 %
f₁ (KHz)	11.7 kHz	13.06 kHz	11.62 %
f₂ (KHz)	62.05 kHz	73.25 kHz	18.05 %
Bandwidth (f₂ – f₁) (KHz)	50.35 kHz	60.19 kHz	19.54 %
f₀/ Q-factor() Bandwidth	0.55	0.46	16.36 %

Table 1.4: The magnitudes of V_R at different frequencies for a low Q circuit

Frequency (f) kHz	Peak Voltage, V_R(V)	I_{peak}(mA)	
1	400 mV	1.99 mA	
20	2.96 V	14.79 mA	
25	3 V	14.99 mA	
26	3.02 V	15.09 mA	
27	3.02 V	15.09 mA	
27.7	3.02 V	15.09 mA	
28	3.02 V	15.09 mA	
30	2.96 V	14.79 mA	
40	2.96 V	14.79 mA	
60	2.8 V	13.99 mA	
80	2.6 V	12.99 mA	
100	2.36 V	11.79 mA	
120	2.12 V	10.59 mA	
140	2 V	9.99 mA	
160	1.8 V	8.99 mA	
180	1.68 V	8.39 mA	
200	1.56 V	7.79 mA	

Table 1.5: Resonant frequency, Quality Factor and Bandwidth for a low Q circuit

	Theoretical, (a)	Experimental, (b)	Deviation (%), $\frac{a-b}{a} \times 100$
f_0	27.7 kHz	27.7 kHz	0%
f_1	6.8 kHz	8.3 kHz	22.1 %
f_2	107.14 kHz	118.68 kHz	10.8 %
Bandwidth ($f_2 - f_1$)	100.34 kHz	110.4 kHz	10 %
Q-factor($\frac{f_0}{\text{Bandwidth}}$)	0.27	0.25	7.4 %

Questions

1. Explain why the load voltage in an RLC circuit is maximum at resonance condition.
2. If a 5mH inductor was used instead of the 330 μ H one, what capacitance value would be required to keep the resonant frequency (f_0) the same as the value you obtained from your experiment?
3. How would the resonant frequency change if the 100 Ω resistor was replaced with a 50 Ω one?
Explain your answer.
4. Use your experimental results and the graphs you obtained from the simulations to explain the concept of high and low Quality Factor in series RLC circuits.
5. Do the practical values of resonant frequency, bandwidth and quality factor you obtained confirm the theoretical values?